PORTFOLIO TEMPLATE

Computer Science

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Period 1 – Year 1

Make a nice frontpage

In Microsoft Word: go to References > Table of Contents to insert or update an automatic table of contents.

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Introduction

Use this section to introduce yourself, your study programme, your learning goals and what the reader can expect in this portfolio and Github name

Setup Guides

Use this chapter to explain how your development environment is installed and configured. Add screenshots where helpful.

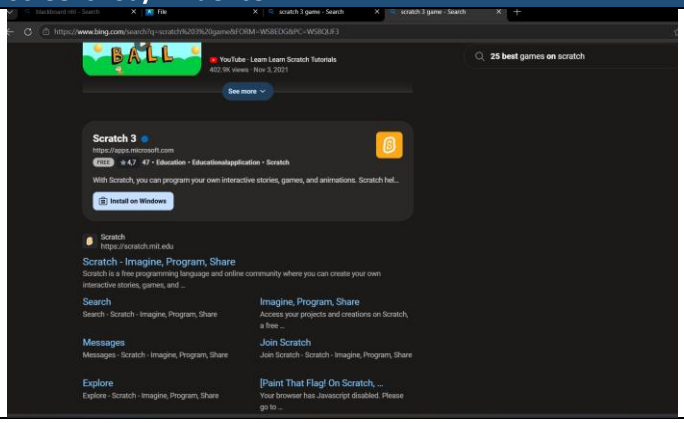
Visual Studio Code installation

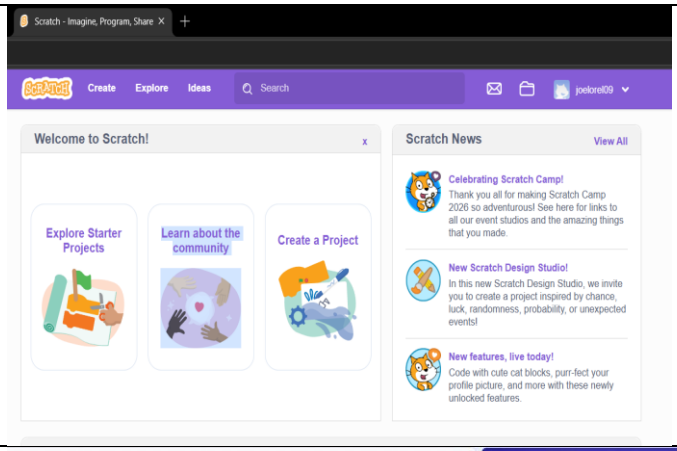
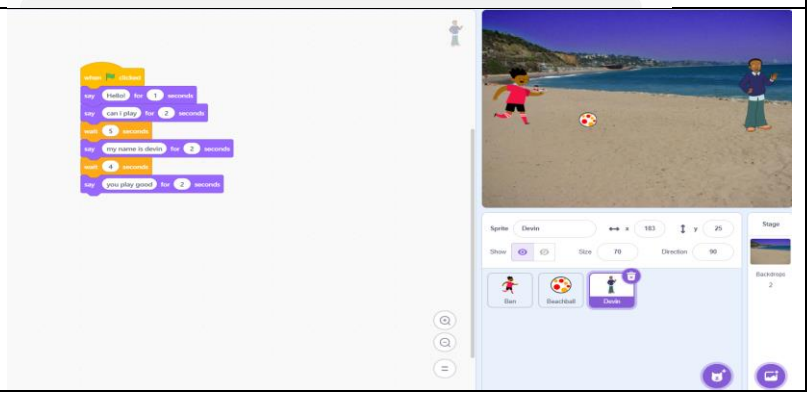
Step	Action	Screenshot / Evidence
1 download vs code on your browser	Describe the action here.	Insert screenshot or link.
2 install vs code	Describe the action here.	Insert screenshot or link.
3 open vs code	Describe the action here.	Insert screenshot or link.

GitHub account and repository setup

Step	Action	Screenshot / Evidence
1 Opening github on browser	Describe the action here.	Insert screenshot or link.
2 create an account	Describe the action here.	Insert screenshot or link.
3 upload your work	Describe the action here.	Insert screenshot or link.

CS50 Codespace setup

Step	Action	Screenshot / Evidence
1 scratch	Open your browser and type scratch 3, then you will see it appear on the screen. Then click	

2 create an account	To create an account, click the create button and enter your credentials	
3 build a game	To build a game, use the blocks that available on the right side of the screen to make a block of codes that will run on the commands you give	

PlatformIO setup for Arduino

Step	Action	Screenshot / Evidence
1 download Arduino ide	Describe the action here.	Insert screenshot or link.
2 make a code	Describe the action here.	Insert screenshot or link.
3 upload to microcontroller	Describe the action here.	Insert screenshot or link.

Computer Science

Use this chapter for Computer Science assignments, code explanations, test results and links to submissions or repositories.

Week 0 Hello, me

A basic first C program prints a message to the terminal. This can be used as an example for documenting code in the portfolio.

```
#include <stdio.h>

int main(void)
{
    printf("Hello, world!\n");
    return 0;
}
```

https://submit.cs50.io/check50/e6443be4a18a376b77478901b3e189d50d48c4eb

Week 1

Section	Content
Assignment	Scratch project(building a scratch game)
Approach	You download the scratch app on you computer, then open it, then create an account and login. First, i created the three sprites, then added a background, then the logic(movement of the ball).
Testing	Add test results, screenshots or check50 output.
Reflection	What did you learn and what would you improve?

Week 2

Section	Content
Assignment	Describe the assignment and requirements.
Approach	Explain your plan, choices and important code logic.
Testing	Add test results, screenshots or check50 output.
Reflection	What did you learn and what would you improve?

Et cetera …

Git Together Assignment

Git url

Git code

Group work explanation

Embedded Software

Use this chapter for Arduino, microcontroller and hardware-related assignments. Include schematics, I/O lists, code and test evidence.

Mario Uno

Mario Uno converts the CS50 Mario More assignment to run on the Arduino Uno through the serial monitor.

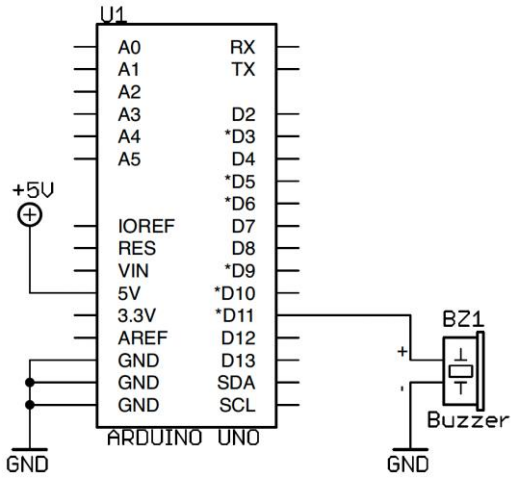
Requirements

The program asks for a pyramid height between 1 and 8 and prints the Mario-style pyramid in the serial monitor. Extra features, such as using the LED matrix, can be added for a higher rating.

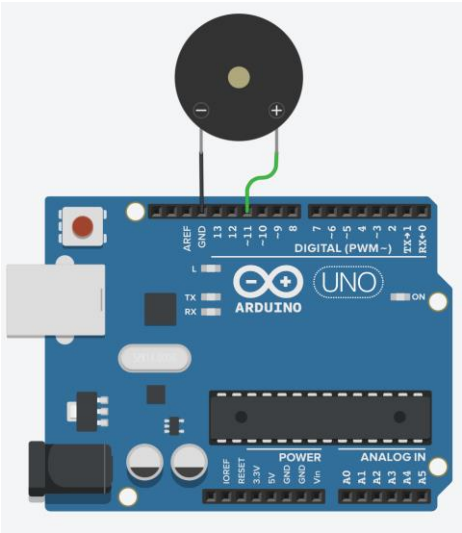
Bill of materials

Component	Purpose	Quantity
Arduino Uno R4 WiFi	Microcontroller board	1
Passive Buzzer	Make the dash and dot audiable	1
Red LED	Make the dash and dot visual	1
Resistor 220 ohm / 0.25W	Current limiter for the LED	1

Schematic



<- OR ->



Code

```
void setup()
{
    Serial.begin(9600);
}

void loop()
{
    // Ask the user for a height between 1 and 8.
    // Print the Mario pyramid to the serial monitor.
}
```

Evidence

Insert a screenshot of the serial monitor output and, if used, a photo or video still of the LED matrix.

Morse Code Generator

Section	Content
Goal	Describe what the project should do.
Components	List all hardware components.
I/O list	Add pin numbers, direction and descriptions.
Code explanation	Explain the most important functions and logic.
Testing and evidence	Add photos, screenshots and results.

Et cetera …

Self Reflection

Use this chapter to reflect on your learning process, planning, collaboration, problem solving and professional development.

Week	What went well?	What was difficult?	Next improvement
Week 1			
Week 2			
Week 3			
Week 4			
Week 5			
Week 6			
Week 7			

Assignment Cards

Use this chapter for signed assignment cards, teacher feedback and assessment evidence. Take a picture of a scan of the assignment cards and put them here

Appendices

Add extra screenshots, diagrams, code fragments, links or feedback here.